Prabhas Teja Penugonda

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EDUCATION

University of Maryland, Baltimore County — **Baltimore, MD**
Master of Professional Studies in Data Science — GPA: 4.0 / 4.0 — *Jan 2025 – Dec 2026*
Relevant coursework: Machine Learning & Data Analysis, Big Data Processing, Data Management, Statistical Modeling, Ethical Issues in Data Science

Jawaharlal Nehru Technological University — **Hyderabad, India**
Bachelor of Technology in Computer Science — *Dec 2021 – Jul 2024*

TECHNICAL SKILLS

Programming Languages: Python, SQL (PostgreSQL, MySQL, SQL Server, T-SQL)
Data Science & Machine Learning: Scikit-learn, XGBoost, TensorFlow, PyTorch, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis, Model Evaluation, Hyperparameter Tuning, SMOTE
Machine Learning Algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Isolation Forest, LSTM Autoencoder, Convolutional Neural Networks (ResNet9)
Data Processing & Engineering: Pandas, NumPy, PySpark (Structured Streaming), Apache Kafka, ETL Pipeline Design, Batch & Stream Processing, Docker
Cloud & Big Data: AWS S3, AWS Glue, Amazon Athena, IAM, CloudShell
Databases: PostgreSQL, MySQL, SQL Server, SQLite
Data Visualization & BI: Power BI, Streamlit, Matplotlib, Seaborn, Plotly
Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code

EXPERIENCE

Parsolvate LLC — **USA**
Data Science Intern — *Apr 2026 – Present*

- Developed and optimized end-to-end machine learning pipelines for classification and regression use cases, improving model accuracy by 18% through feature engineering, hyperparameter tuning, and cross-validation using Python, Scikit-learn, and XGBoost.
- Designed SQL-based data transformation workflows and ETL pipelines to process datasets containing 1M+ records, reducing data preparation time by 60% through automated cleansing, validation, and aggregation.
- Built interactive Power BI dashboards and automated reporting solutions to monitor 20+ business and model performance KPIs, reducing manual reporting effort by 70% and enabling faster data-driven decision-making.

CodTech IT Solutions — **Hyderabad, India**
Data Science Intern — *Aug 2024 – Oct 2024*

- Performed end-to-end EDA on a 400K+ record US used car dataset (Pandas, NumPy, Matplotlib, Seaborn); built IQR-based outlier removal and missing-value imputation pipelines to prepare data for modeling
- Engineered features with one-hot encoding and trained a linear regression model for price prediction; evaluated performance with MAE, MSE, and R2
- Built 15+ statistical visualizations (heatmaps, scatter plots, distributions) to communicate pricing trends across 40+ manufacturers and vehicle conditions to stakeholders

PROJECTS

Real-Time Bitcoin Anomaly Detection Pipeline ***/ Apache Kafka, PySpark, Docker, Isolation Forest, LSTM Autoencoder, Streamlit*** **| GitHub**

- Designed and built a production-style streaming pipeline: Kraken WebSocket → Apache Kafka → PySpark Structured Streaming → live Streamlit dashboard; sustained 0.75 trades/sec throughput in 5-second micro-batch windows.
- Deployed dual ML anomaly detection: Isolation Forest (online retraining every 500 ticks, avg score 0.52) and LSTM Autoencoder (adaptive threshold mean + 2.5σ); achieved a 7.2% anomaly rate across live BTC/USD data.
- Containerized all 5 services with Docker Compose; layered 4 statistical detectors (3σ volatility, 2σ mean reversion, spike, jump) with severity-based alerting (44% HIGH / 33% MED / 22% LOW); added WebSocket auto-reconnect and Kafka retry logic for fault tolerance.

Consumer Complaints Cloud Analytics Pipeline ***/ AWS S3, Glue, Athena, Power BI, Python*** **| GitHub**

- Built a cloud analytics pipeline ingesting 500K CFPB complaint records: converted 8.3 GB CSV to Parquet/Snappy (4.7× compression), stored in Amazon S3, cataloged with AWS Glue, and queried serverlessly via Amazon Athena; wrote 5 advanced SQL queries using LAG, ROW_NUMBER PARTITION BY, and cumulative SUM OVER window functions.
- Connected Athena to Power BI via ODBC; built a 4-page dashboard with AI-powered Key Influencers visual (Debt collection identified as 9.6× untimely-response driver), Decomposition Tree drill-down, and product × state heatmap matrix.

Public Health Risk Prediction — CDC BRFSS ***/ Python, Scikit-learn, XGBoost, SMOTE*** **| GitHub**

- Applied CRISP-DM workflow to CDC BRFSS LLCP2024 survey data to identify key diabetes risk drivers; addressed severe class imbalance using SMOTE before model training.
- Compared Logistic Regression, Random Forest, and Gradient Boosting; Gradient Boosting achieved ROC-AUC 0.758 on held-out test set with age, BMI, and cardiovascular indicators as top features.
- Ranked feature importance across models to surface top-5 risk indicators; framed findings as screening-prioritization recommendations for public health decision-making.

Crop Disease Classification — ResNet9 CNN ***/ Python, PyTorch, SSIS, SQL Server, Power BI*** **| GitHub**

- Trained a ResNet9 CNN on 87,000+ labeled leaf images across 38 crop disease categories; achieved 85% test accuracy using data augmentation and stratified train/validation splits.
- Built a 4-layer pipeline: CNN inference → SSIS ETL → normalized SQL Server schema → Power BI dashboards; deployed inference via a Gradio web application with versioned model weights on GitHub Releases.
- Authored T-SQL analytical queries and schema definitions; built Power BI dashboards surfacing disease distribution, severity risk, and class frequency across crop types.

LEADERSHIP & ACTIVITIES

JNTUH Technical Fest — **Hyderabad, India**
Coding Contest Organizer — ***2022 – 2024***

- Organized and conducted coding competitions across two annual technical fests; coordinated 200+ participants from multiple colleges including logistics, judging, and workshop facilitation.

National Service Scheme (NSS) — **Hyderabad, India**
Volunteer — ***2022 – 2024***

- Led community outreach initiatives including sanitation improvement drives and education awareness campaigns in government schools focused on female education.